Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A dielectric slab of thickness d is inserted in a parallel plate capacitor whose negative plate is at $x = 0$ and positive plate is at $x = 3d$. The slab is equidistant from the plates. The capacitor is given some charge. As x goes from 0 to $3d$,

(A) the magnitude of the electric field remains the same

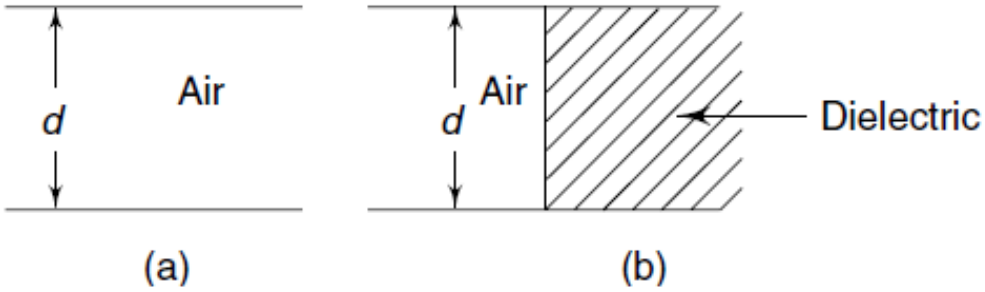
(B) the direction of the electric field changes continuously

(C) the electric potential increases continuously

(D) the electric potential increases at first, then decreases and again increases.

Q.2

A parallel plate air filled capacitor shown in Fig.(a) has a capacitance of $2\mu\text{F}$. When it is half filled with a dielectric of dielectric constant $k = 3$ as shown in Fig.(b), its capacitance becomes



(A) $4\mu\text{F}$

(B) $3\mu\text{F}$

(C) $1.5\mu\text{F}$

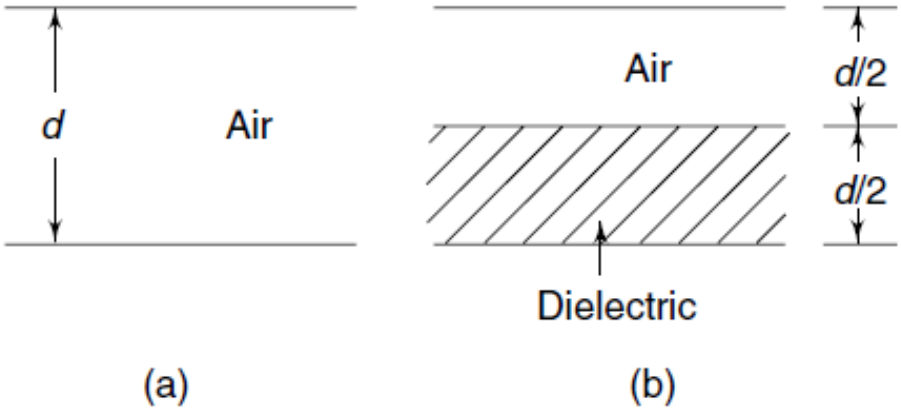
(D) $0.5\mu\text{F}$

Q.3

A parallel plate air filled capacitor shown in Fig. (a) has a capacitance of $2\mu\text{F}$. When it is half filled with a

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dielectric of dielectric constant $k = 3$ as shown in Fig. (b), its capacitance becomes



(A) $\frac{1}{3}\mu\text{F}$

(B) $1\mu\text{F}$

(C) $3\mu\text{F}$

(D) $9\mu\text{F}$

Q.4

A parallel plate capacitor of capacitance 100pF is to be constructed by using paper sheets of $1.0\ \text{mm}$ thickness as dielectric. If the dielectric constant of paper is 4.0, the number of circular metal foils of diameter $2.0\ \text{cm}$ each required for this purpose is

(A) 10

(B) 20

(C) 30

(D) 40

Q.5

A $2\mu\text{F}$ capacitor C_1 is charged to a voltage $100\ \text{V}$ and a $4\mu\text{F}$ capacitor C_2 is charged to a voltage $50\ \text{V}$. The capacitors are then connected in parallel. What is the loss of energy due to parallel connection?

(A) $1.7\ \text{J}$

(B) $1.7 \times 10^{-1}\ \text{J}$

(C) $1.7 \times 10^{-2}\ \text{J}$

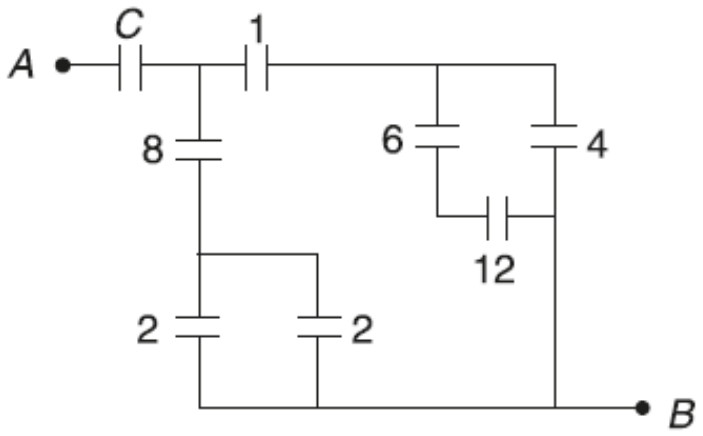
(D) $1.7 \times 10^{-3}\ \text{J}$

Q.6

Figure shows a network of capacitors where the numbers indicate capacitances in microfarad. What must be the

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value of capacitance C if the equivalent capacitance between points A and B is to be $1\mu\text{F}$?



(A) $\frac{31}{23}\mu\text{F}$

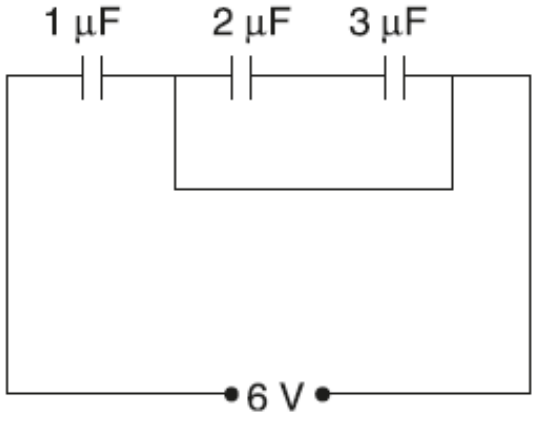
(B) $\frac{32}{23}\mu\text{F}$

(C) $\frac{33}{23}\mu\text{F}$

(D) $\frac{34}{23}\mu\text{F}$

Q.7

Figure shows three capacitors connected to a $6\ \text{V}$ power supply. What is the charge on the $2\mu\text{F}$ capacitor?



(A) $1\mu\text{C}$

(B) $2\mu\text{C}$

(C) $3\mu\text{C}$

(D) $4\mu\text{C}$

Q.8

Two identical capacitors were initially connected in parallel and both of them acquired a common potential V. If these capacitors are then separated and reconnected but this time in series, with the positive plate of one being connected to the negative of the other, then choose the true statement out of the following.

(A) The charges on the free plate connected together are destroyed.

(B) The energy stored in the system increases.

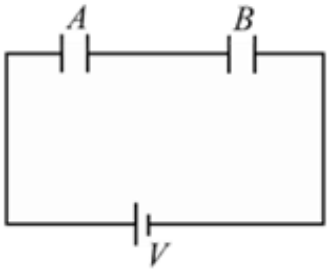
(C) The potential difference between the free plates is $2\ \text{V}$.

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(D) The potential difference remains constant.

Q.9

Two identical parallel plate capacitors A and B are connected in series through a battery of potential difference V (see figure). Area of each plate is a and initially plates of capacitors are separated by a distance d. Now, separation between plates of capacitor B starts increasing at constant rate v, find the rate by which work is done on the battery when separation between plates of capacitor B is $2d$.



(A) $\frac{a\varepsilon_0 v V^2}{d^2}$

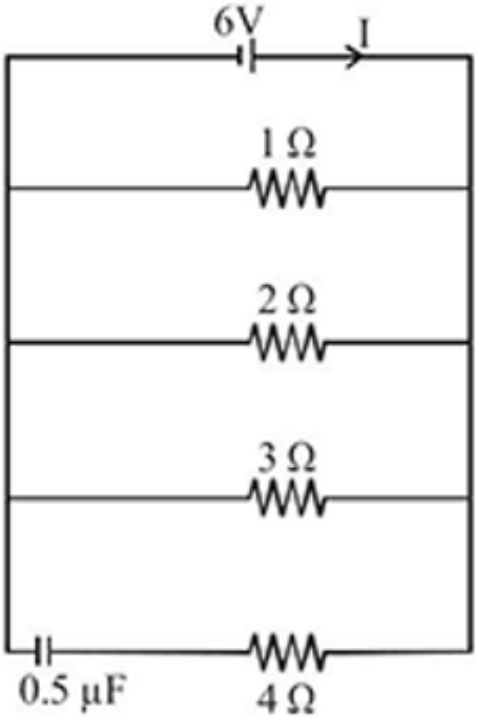
(B) $\frac{a\varepsilon_0 v V^2}{2d^2}$

(C) $\frac{a\varepsilon_0 v V^2}{9d^2}$

(D) $\frac{2a\varepsilon_0 v V^2}{d^2}$

Q.10

In the given circuit diagram the current through the battery and the charge on the capacitor respectively in steady state are



(A) 1 Aand $3\mu C$

(B) 17 A and $0\mu C$

(C) $\frac{6}{7}$ A and $\frac{12}{7}\mu C$

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(D) 11 A and 3μC

Q.11

Capacitance of a capacitor becomes $\frac{4}{3}$ times its original value if a dielectric slab of thickness $t = d/2$ is inserted between the plates (d = separation between the plates). The dielectric constant of the slab is

(A) 2

(B) 4

(C) 6

(D) 8

Q.12

Two infinitely long parallel conducting plates having surface charge densities $+\sigma$ and $-\sigma$ respectively, are separated by a small distance. The medium between the plates is vacuum. If ε_0 is the dielectric permittivity of vacuum, then the electric field in the region between the plates is

(A) zero

(B) $\frac{\sigma}{2\varepsilon_0}$ volts/meter

(C) $\frac{\sigma}{\varepsilon_0}$ volts/meter

(D) $\frac{2\sigma}{\varepsilon_0}$ volts/meter

Q.13

A parallel plate capacitor is maintained at a certain potential difference. When a dielectric slab of thickness 3 mm is introduced between the plates, the plate separation had to be increased by 2 mm in order to maintain the same potential difference between the plates. The dielectric constant of the slab is

(A) 2

(B) 3

(C) 4

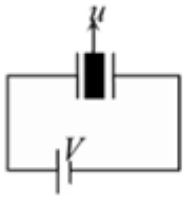
(D) 5

Q.14

A parallel plate capacitor having square plate of area a and separation between plates d is completely filled with a dielectric of dielectric constant k. Capacitor is connected with a battery of potential difference V. Now

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dielectric is pulled with constant speed u. Find the initial value of current.



(A) $\frac{\varepsilon_0\sqrt{au}}{2d}(k-1)$

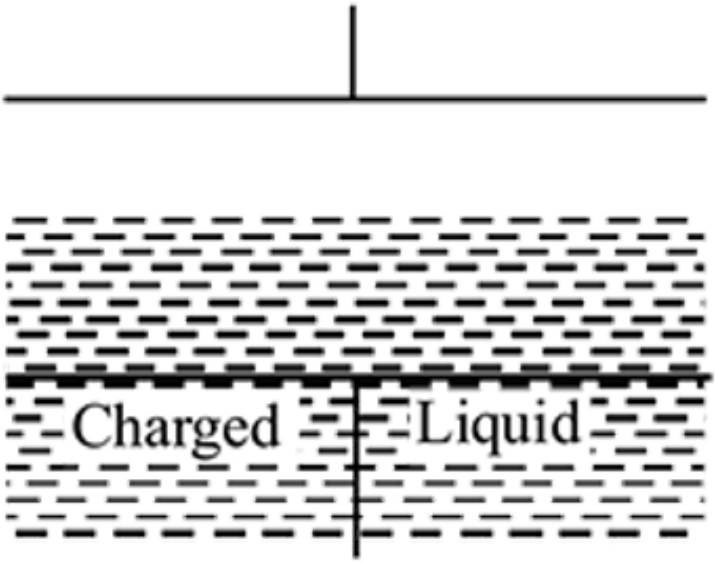
(B) $\frac{\varepsilon_0\sqrt{au}}{d}(k-1)$

(C) zero

(D) $\frac{2\varepsilon_0\sqrt{au}}{d}(k-1)$

Q.15

A parallel plate capacitor is located horizontally such that one of the plates is submerged in a liquid while the other is above the liquid surface. When plates are charged the level of liquid



(A) rises

(B) falls

(C) remains unchanged

(D) may rise or fall depending on the of charge amount

Answers & Solutions

Q.1 Answer:

the electric potential increases continuously

Solution:

The insertion of the dielectric slab decreases the electric field without changing its direction. The electric potential increases as we go from the negative to the positive plate. Hence the correct choice is (c).

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Q.2 Answer:

$4\mu\text{F}$

Solution:

If A is the area of each plate, the capacitance of the air-filled capacitor shown in Fig. 11.53(a) on page 11.26 is

$C_0 = \frac{\varepsilon_0 A}{d},$ where $C_0 = 2\mu\text{F}$ (given).

The capacitance of air capacitor in Fig. 11.54 (b) is

$C_1 = \frac{\varepsilon_0 A/2}{d} = \frac{\varepsilon_0 A}{2d} = \frac{C_0}{2}$

The capacitance of dielectric filled capacitor in Fig. $11.53(\text{ b})$ is

$C_2 = \frac{k\varepsilon_0 A/2}{d} = \frac{k\varepsilon_0 A}{2d} = \frac{kC_0}{2}$

Since C_1 and C_2 are in parallel, the capacitance C of the capacitor shown in Fig. (b) is

$$C = C_1 + C_2 = \frac{C_0}{2} + \frac{kC_0}{2}$$
$$= \frac{C_0}{2}(1+k) = \frac{2\mu\text{F}}{2}(1+3) = 4\mu\text{F}$$

Hence the correct choice is (a).

Q.3 Answer:

$3\mu\text{F}$

Solution:

If A is the area of each plate, the capacitance of the air capacitor shown in Fig. (a) is

$C_0 = \frac{\varepsilon_0 A}{d},$ where $C_0 = 2\mu\text{F}$ (given).

The capacitance of air capacitor in Fig. (b) is

$C_1 = \frac{\varepsilon_0 A}{d/2} = \frac{2\varepsilon_0 A}{d} = 2C_0$

The capacitance of the dielectric filled capacitor in Fig. (b) is

$C_2 = \frac{k\varepsilon_0 A}{d/2} = \frac{2k\varepsilon_0 A}{d} = 2kC_0$

where k is the dielectric constant. Now capacitors C_1 and C_2 are in series. Therefore, the capacitance C of the capacitor shown in Fig. 11.52 (b) is given by

$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{2C_0} + \frac{1}{2kC_0} = \frac{(k+1)}{2C_0 k}$

Or $C = \frac{2C_0 k}{(k+1)} = \frac{2\times 2\mu\text{F}\times 3}{(3+1)} = 3\mu\text{F}$

Hence the correct choice is (c).

Q.4 Answer:

10

Solution:

$C = 100\text{pF} = 100 \times 10^{-12}\ \text{F} = 10^{-10}\ \text{F}$

Let the number of sheets of foils required be n. They will form $(n-1)$ capacitors. If K is the dielectric constant of the dielectric, the capacitance is given by

$C = \frac{K\varepsilon_0 (n-1) A}{d}$

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$$\text{or} \quad n-1=\frac{Cd}{K\varepsilon_0 A}=\frac{Cd}{K\cdot 4\pi\varepsilon_0}\cdot\frac{4\pi}{\pi r^2}$$

$$=\frac{4Cd}{K\cdot 4\pi\varepsilon_0 r^2}$$

$$=\frac{4\times 10^{-10}\times 1\times 10^{-3}\times 9\times 10^9}{4\times(1.0\times 10^{-2})^2}=9$$

$$\text{or } n=10$$

Hence the correct choice is (a).

Q.5 Answer:

1.7×10^{-3} J

Solution:

Charge Q_1 and $C_1=C_1V_1=2\times 10^{-6}\times 100=2\times 10^{-4}$C.
Charge Q_2 on $C_2=C_2V_2=4\times 10^{-6}\times 50=$
2×10^{-4}C. Total charge $Q=Q_1+Q_2=4\times 10^{-4}$C.
Total energy before connection is

$$E_1=\frac{1}{2}C_1V_1^2+\frac{1}{2}C_2V_2^2$$

$$=\frac{1}{2}\times 2\times 10^{-6}\times(100)^2+\frac{1}{2}\times 4\times 10^{-6}\times(50)^2$$

$$=1.5\times 10^{-2}\text{ J}$$

The common potential difference V after connection is given by

$C_1V+C_2V=Q$

or $V=\frac{Q}{C_1+C_2}$

Therefore, total energy after connection is

$$E_2=\frac{1}{2}(C_1+C_2)V^2=\frac{1}{2}\times\frac{Q^2}{(C_1+C_2)}$$

$$=\frac{1}{2}\times\frac{\left(4\times 10^{-4}\right)^2}{(2+4)\times 10^{-6}}=1.33\times 10^{-2}\text{ J}$$

$\therefore$ **Loss of energy $=E_1-E_2=0.17\times 10^{-2}$ J. Hence the correct choice is (d).**

Q.6 Answer:

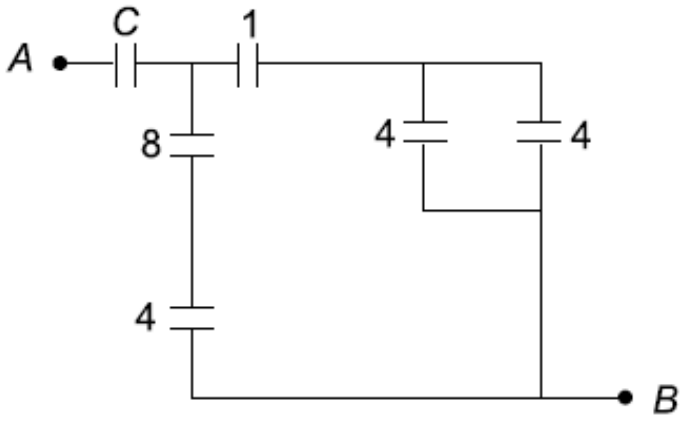
$\frac{32}{23}\mu$F

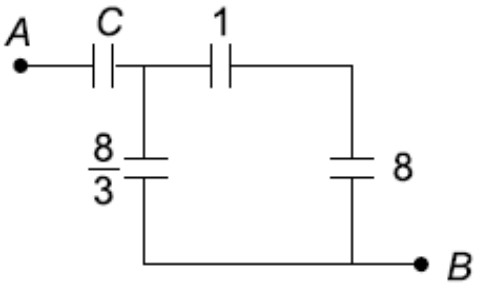
Solution:

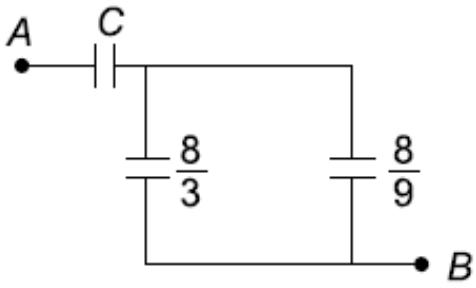
The series combination of 6 and 12 is equivalent to 4 and the parallel combination of 2 and 2 is also equivalent to 4. Therefore the network can be simplified as shown in Fig. The parallel combination of 4 and 4 is equivalent to 8 and the series combination of 8 and 4 is equivalent to $8/3$. Thus the combination in Fig. reduces to that in Fig. The series combination of 1 and 8 in Fig. yields $8/9$ as

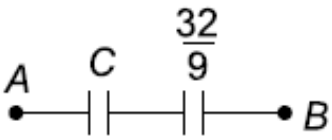
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shown in Fig.









Now $8/3$ and $8/9$ are in parallel and their equivalent is $32/9$. Therefore, the network finally reduces to that in Fig. Since the total capacitance between A and B is to be (i.e. $1\mu\text{F}$), we have $1 = \frac{1}{C} + \frac{9}{32}$

$\Rightarrow \quad C = \frac{32}{23}\mu\text{F}$. Hence the correct choice is (b).

Q.7 Answer:

$2\mu\text{C}$

Solution:

Capacitors of capacitances $2\mu\text{F}$ and $3\mu\text{F}$ are in parallel and this combination is in series with $1\mu\text{F}$ capacitor. Thus, we have $1\mu\text{F}$ capacitor in series with $5\mu\text{F}$ capacitor and the potential difference across this series combination is $6\ \text{V}$. Therefore, the potential differences across $5\mu\text{F}$ capacitor (which consists of a parallel combination of $2\mu\text{F}$ and $3\mu\text{F}$ capacitors) is $1\ \text{V}$. Hence the charge on $2\mu\text{F}$ capacitor $= 2\mu\text{F} \times 1\ \text{V} = 2\mu\text{C}$, which is choice (b).

Q.8 Answer:

The potential difference between the free plates is 2 V.

Solution:

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When the two capacitors charged to same potential are connected in series, then total potential difference

$V^{'} = V_1 + V_2 = V + V = 2\text{V}$

Q.9 Answer:

$\frac{a\varepsilon_0 v V^2}{9d^2}$

Solution:

Let at any instant separation between plates of capacitor B is x, then

$C_{eq} = \frac{\frac{a\varepsilon_0}{d} \cdot \frac{a\varepsilon_0}{x}}{\frac{a\varepsilon_0}{d} + \frac{a\varepsilon_0}{x}} = \frac{a\varepsilon_0 (dx)}{dx(d+x)} = \frac{a\varepsilon_0}{d+x}$

$Q = C_{eq} V = \frac{a\varepsilon_0}{d+x} V$

$\frac{dQ}{dt} = -\frac{a\varepsilon_0}{(d+x)^2} V \frac{dx}{dt} = -\frac{a\varepsilon_0}{(d+x)^2} V v$

Rate of work done on the battery $= -\left(\frac{dQ}{dt}\right) V = \frac{a\varepsilon_0 v V^2}{9d^2}$ **(when** $x = 2d$**)**

Q.10 Answer:

$11\ \text{A}$ and $3\mu\text{C}$

Solution:

In steady state capacitor is fully charged and no current flows through it.

$\frac{1}{\text{R}_{\text{eq}}} = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} = \frac{6+3+2}{6} = \frac{11}{6} \Rightarrow \text{R}_{\text{eq}} = \frac{6}{11}\Omega$

Current through battery $= \frac{6 \times 11}{6} = 11\ \text{A}$

$\text{Q} = \text{CV} = 0.5 \times 10^{-6} \times 6 = 3.0 \times 10^{-6}\text{C} = 3\mu\text{C}$

Q.11 Answer:

2

Solution:

$C = \frac{\varepsilon_0 A}{d}, \quad \frac{4C}{3} = \frac{\varepsilon_0 A}{\left[d - \frac{d}{2} + \frac{d}{2k}\right]}, \frac{4}{3} = \frac{d}{\frac{d}{2} + \frac{d}{2k}}, 2d + \frac{2d}{k} = 3d, \quad k = 2$

Q.12 Answer:

$\frac{\sigma}{\varepsilon_0}$ volts/meter

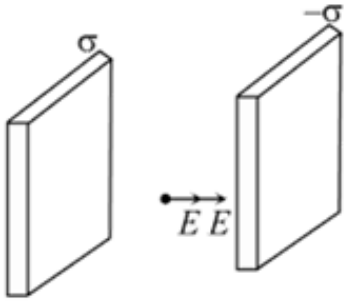
Solution:

Electric field between the plates is

$= \frac{\sigma}{2\varepsilon_0} - \frac{(-\sigma)}{2\varepsilon_0}$

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$= \frac{\sigma}{\varepsilon_0}$ **volt/meter**



Q.13 Answer:

3

Solution:

$V' = \frac{Q}{C^1} = \frac{Q\left[d' - t\left(1 - \frac{1}{K}\right)\right]}{\varepsilon_0 A}, \quad V = \frac{Q}{C} = \frac{Qd}{\varepsilon_0 A}$

Now, $V' = V \Rightarrow d = d' - t\left(1 - \frac{1}{K}\right)$

$d' - d = 2\ \text{mm}, t = 3\ \text{mm} \quad \Rightarrow \quad 2 = 3\left(1 - \frac{1}{K}\right) \Rightarrow K = 3$

Q.14 Answer:

$\frac{\varepsilon_0 \sqrt{a} u}{d}(k-1)$

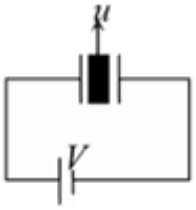
Solution:

$C = \frac{k\sqrt{a}(\sqrt{a}-x)\varepsilon_0}{d} + \frac{\sqrt{a}x\varepsilon_0}{d}$

$Q = CV$

$I = \left|\frac{dQ}{dt}\right| = \left|V\frac{dC}{dt}\right| = \left|\left[\frac{-k\sqrt{a}\varepsilon_0}{d}u + \frac{\sqrt{a}\varepsilon_0}{d}u\right]\right|$

$= \frac{\varepsilon_0\sqrt{a}u}{d}(k-1)$



Q.15 Answer:

rises

Solution:

The molecules of liquid will convert into induced dipole, get oriented along the electric field produced between the plates and rise due to force of attraction

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